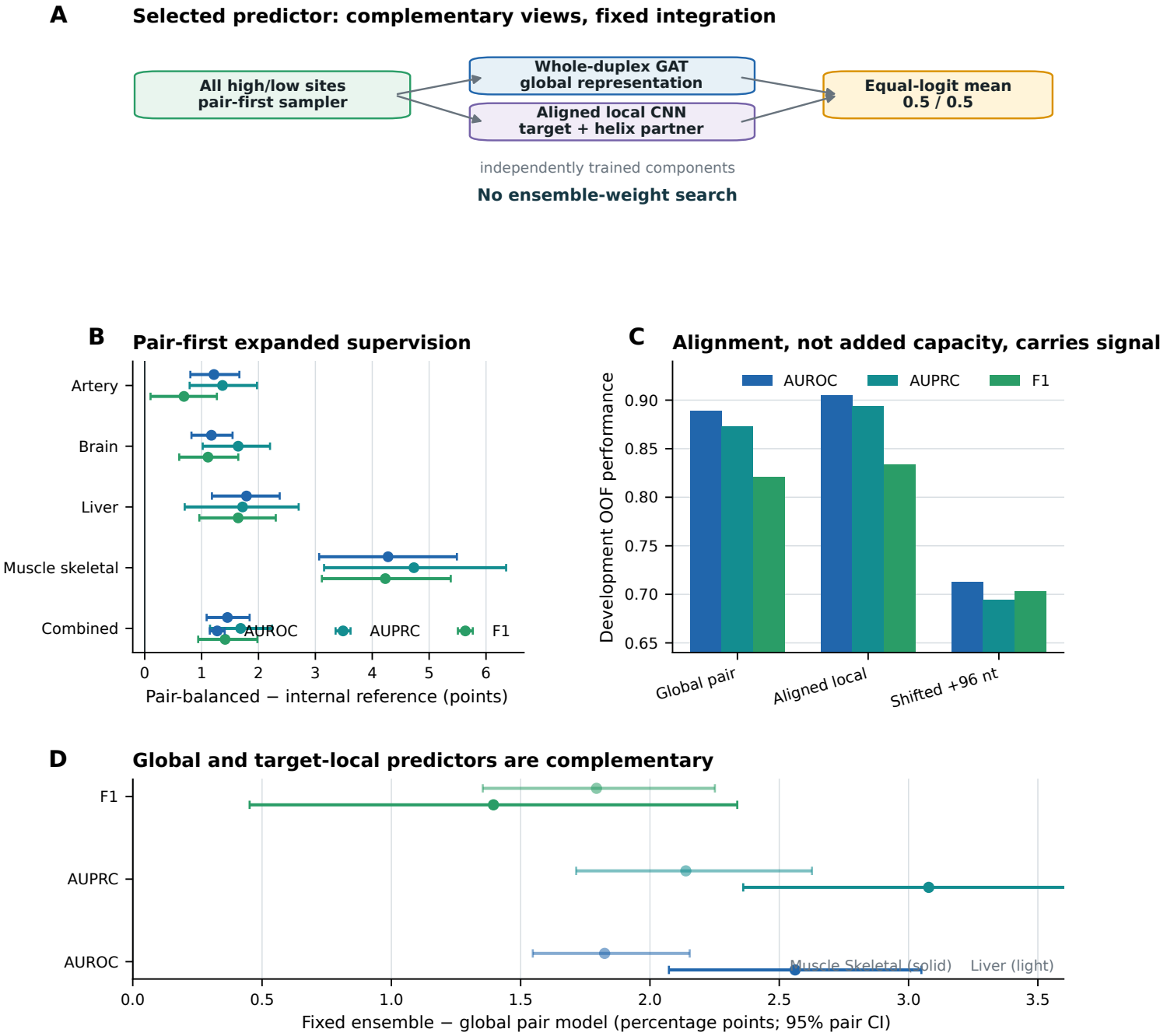
A
Selected predictor: complementary views, fixed integration
All high/low sites pair-first sampler
Whole-duplex GAT global representation
Aligned local CNN target + helix partner
Equal-logit mean 0.5 / 0.5
independently trained components
No ensemble-weight search
B
Pair-first expanded supervision
Artery
Brain
Liver
Muscle skeletal
Combined
AUROC
AUPRC
F1
0
1
2
3
4
5
6
Pair-balanced − internal reference (points)
C
Alignment, not added capacity, carries signal
AUROC
AUPRC
F1
Development OOF performance
0.90
0.85
0.80
0.75
0.70
0.65
Global pair
Aligned local
Shifted +96 nt
D
Global and target-local predictors are complementary
F1
AUPRC
AUROC
Muscle Skeletal (solid)
Liver (light)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Fixed ensemble − global pair model (percentage points; 95% pair CI)